NEET UG PHYSICS | MECHANICS

Work, Energy and Power

Comprehensive Concept Tutorial, Worked Illustrations, and NEET UG Practice Assessment

1. Learning Objectives & Chapter Overview

In everyday language, the terms *work*, *energy*, and *power* are used loosely. A farmer ploughing a field, a construction worker carrying bricks, a student studying for a competitive examination, or an artist painting a landscape are all said to be working. Similarly, someone with the stamina to work 14–16 hours a day is said to have large energy, and fast punches in boxing or karate are described as powerful. In physics, however, these three quantities have exact, quantitative definitions that only loosely correlate with physiological effort.

Key Competencies for NEET UG

- Evaluate the **scalar (dot) product** of two vectors in geometric and Cartesian component forms, including vector projections.
- Apply the **Work-Energy (WE) Theorem** for both constant and spatially varying forces.
- Distinguish between positive, negative, and zero work, and calculate work from the area under a force-displacement $F(x)$–x graph.
- Formulate **potential energy** $V(x)$ for conservative forces via $F(x) = -\dfrac{\mathrm{d}V}{\mathrm{d}x}$ and apply conservation of mechanical energy to free fall, vertical circular motion, and ideal springs obeying Hooke's law.
- Analyze **average and instantaneous power** $P = \mathbf{F} \cdot \mathbf{v}$ and unit conversions (W, hp, kWh).
- Solve **elastic, inelastic, and completely inelastic collisions** in one and two dimensions using momentum and kinetic energy principles.

2. Mathematical Prerequisite: The Scalar Product

Physical quantities such as displacement, velocity, acceleration, and force are vectors. While vectors can be added and subtracted, they can also be multiplied in two distinct ways: the **scalar product** (which produces a scalar from two vectors) and the **vector product** (which produces a new vector). For work and power, we require the scalar product.

Definition: Scalar (Dot) Product

The scalar product or dot product of any two vectors $\mathbf{A}$ and $\mathbf{B}$, denoted as $\mathbf{A} \cdot \mathbf{B}$ (read "A dot B"), is defined as:

$$\mathbf{A} \cdot \mathbf{B} = AB\cos\theta$$

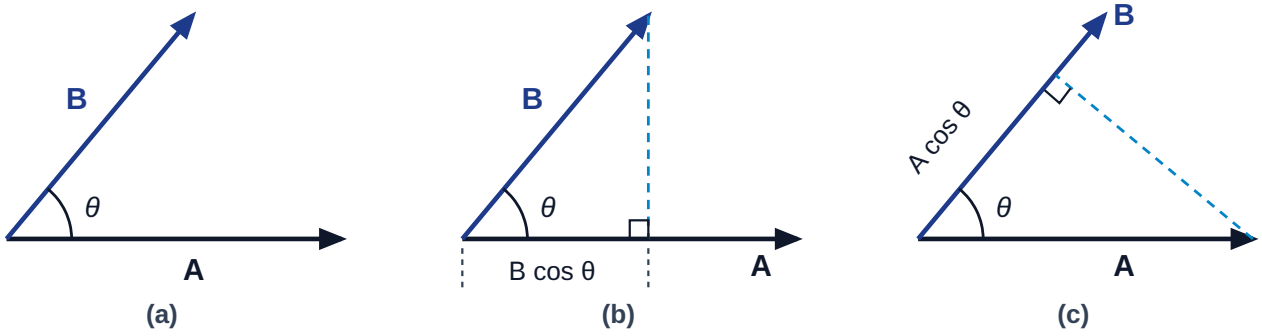
where $A = |\mathbf{A}|$ and $B = |\mathbf{B}|$ are the magnitudes of the vectors and θ is the angle between them when placed tail-to-tail. Since A, B, and $\cos\theta$ are scalars, $\mathbf{A} \cdot \mathbf{B}$ is a scalar quantity possessing no direction.

Geometric Interpretation & Projections

We can rewrite the defining equation in two equivalent ways:

$$\mathbf{A} \cdot \mathbf{B} = A(B\cos\theta) = B(A\cos\theta)$$

- Geometrically, $B\cos\theta$ is the **projection of $\mathbf{B}$ onto $\mathbf{A}$** (the component of $\mathbf{B}$ along $\mathbf{A}$).
- Similarly, $A\cos\theta$ is the **projection of $\mathbf{A}$ onto $\mathbf{B}$** (the component of $\mathbf{A}$ along $\mathbf{B}$).
- Thus, $\mathbf{A} \cdot \mathbf{B}$ is the product of the magnitude of one vector and the component of the other vector along the direction of the first.



Tutorial Diagram 1: (a) Two vectors A and B inclined at angle θ. (b) B cos θ is the projection of B onto A. (c) A cos θ is the projection of A onto B.

Algebraic Properties & Cartesian Component Form

- **Commutative Law:** $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$
- **Distributive Law:** $\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$
- **Scalar Multiplication:** $\mathbf{A} \cdot (\lambda\mathbf{B}) = \lambda(\mathbf{A} \cdot \mathbf{B})$, where λ is any real number.
- **Orthonormal Unit Vectors ($\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$):**

$$\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = \hat{\mathbf{j}} \cdot \hat{\mathbf{j}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{k}} = 1 \quad (\text{since } \cos 0^\circ = 1)$$
$$\hat{\mathbf{i}} \cdot \hat{\mathbf{j}} = \hat{\mathbf{j}} \cdot \hat{\mathbf{k}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{i}} = 0 \quad (\text{since } \cos 90^\circ = 0)$$

- **Component Form:** For $\mathbf{A} = A_x\hat{\mathbf{i}} + A_y\hat{\mathbf{j}} + A_z\hat{\mathbf{k}}$ and $\mathbf{B} = B_x\hat{\mathbf{i}} + B_y\hat{\mathbf{j}} + B_z\hat{\mathbf{k}}$,
$$\mathbf{A} \cdot \mathbf{B} = A_xB_x + A_yB_y + A_zB_z$$
- **Self Dot Product (Magnitude Squared):**
$$\mathbf{A} \cdot \mathbf{A} = A_x^2 + A_y^2 + A_z^2 = A^2$$
- **Condition for Orthogonality (Perpendicularity):** $\mathbf{A} \cdot \mathbf{B} = 0$ if $\mathbf{A}$ and $\mathbf{B}$ are mutually perpendicular ($\theta = 90^\circ$ or $\pi/2$ rad).

Worked Illustration 1: Angle and Projection Using Scalar Product

Problem: Find the angle between force $\mathbf{F} = (3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 5\hat{\mathbf{k}})$ unit and displacement $\mathbf{d} = (5\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 3\hat{\mathbf{k}})$ unit. Also find the projection of $\mathbf{F}$ on $\mathbf{d}$.

Solution:

First, evaluate the dot product $\mathbf{F} \cdot \mathbf{d}$ using Cartesian components:
$$\mathbf{F} \cdot \mathbf{d} = F_xd_x + F_yd_y + F_zd_z = (3)(5) + (4)(4) + (-5)(3) = 15 + 16 - 15 = 16\,\text{unit}$$
Next, compute the magnitudes of $\mathbf{F}$ and $\mathbf{d}$:
$$F^2 = \mathbf{F} \cdot \mathbf{F} = F_x^2 + F_y^2 + F_z^2 = 3^2 + 4^2 + (-5)^2 = 9 + 16 + 25 = 50\,\text{unit} \implies F = \sqrt{50}\,\text{unit}$$
$$d^2 = \mathbf{d} \cdot \mathbf{d} = d_x^2 + d_y^2 + d_z^2 = 5^2 + 4^2 + 3^2 = 25 + 16 + 9 = 50\,\text{unit} \implies d = \sqrt{50}\,\text{unit}$$
Using $\mathbf{F} \cdot \mathbf{d} = Fd\cos\theta$:
$$\cos\theta = \frac{\mathbf{F} \cdot \mathbf{d}}{Fd} = \frac{16}{\sqrt{50}\sqrt{50}} = \frac{16}{50} = 0.32 \implies \theta = \cos^{-1}(0.32)$$
The projection of $\mathbf{F}$ on $\mathbf{d}$ is given by $F\cos\theta$:
$$\text{Projection of } \mathbf{F} \text{ on } \mathbf{d} = F\cos\theta = \frac{\mathbf{F} \cdot \mathbf{d}}{d} = \frac{16}{\sqrt{50}} = \frac{16}{5\sqrt{2}}\,\text{unit} \approx 2.26\,\text{unit}$$

3. Notions of Work, Kinetic Energy, and the Work-Energy Theorem

Consider rectilinear motion under constant acceleration a. From kinematics, the initial speed u, final speed v, and distance traversed s are related by:
$$v^2 - u^2 = 2as$$
Multiplying both sides by $m/2$ and invoking Newton's Second Law ($F = ma$):
$$\frac{1}{2}mv^2 - \frac{1}{2}mu^2 = mas = Fs$$
Generalising this relation to three dimensions using acceleration vector $\mathbf{a}$ and displacement vector $\mathbf{d}$, we have $v^2 - u^2 = 2\,\mathbf{a} \cdot \mathbf{d}$. Multiplying by $m/2$ yields:
$$\frac{1}{2}mv^2 - \frac{1}{2}mu^2 = m\,\mathbf{a} \cdot \mathbf{d} = \mathbf{F} \cdot \mathbf{d}$$
This equation motivates two fundamental definitions:

- The quantity $\frac{1}{2}mv^2$ is called **kinetic energy** (K). Thus, the left-hand side represents $K_f - K_i$, the change in kinetic energy.
- The quantity $\mathbf{F} \cdot \mathbf{d}$ is called **work** (W) done by the force $\mathbf{F}$ over displacement $\mathbf{d}$.

The Work-Energy (WE) Theorem

Statement: The change in kinetic energy of a particle is equal to the work done on it by the net force acting on it:

$$K_f - K_i = \Delta K = W_{\text{net}}$$

Worked Illustration 2: Raindrop Falling Against Air Resistance

Problem: A raindrop of mass $1.00\,\text{g}$ falls from a height of $1.00\,\text{km}$ under the influence of the downward gravitational force and an opposing resistive force (proportional to speed, but otherwise undetermined). It hits the ground with a speed of $50.0\,\text{m s}^{-1}$. Taking $g = 10\,\text{m s}^{-2}$, find: (a) the work done by the gravitational force, and (b) the work done by the unknown resistive force.

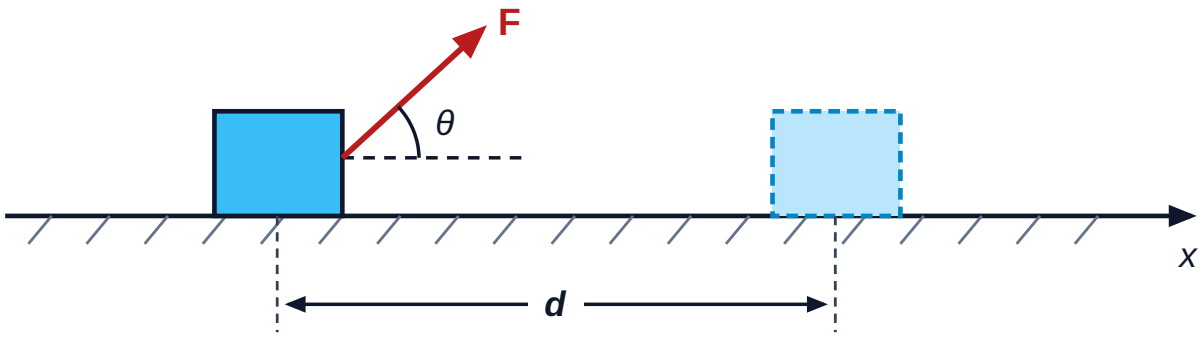
Solution:

Given: $m = 1.00\,\text{g} = 10^{-3}\,\text{kg}$, $h = 1.00\,\text{km} = 10^3\,\text{m}$, initial speed $u = 0$, final speed $v = 50.0\,\text{m s}^{-1}$.

The change in kinetic energy of the drop is:
$$\Delta K = \frac{1}{2}mv^2 - 0 = \frac{1}{2} \times 10^{-3}\,\text{kg} \times (50.0\,\text{m s}^{-1})^2 = 1.25\,\text{J}$$
(a) Work done by the downward gravitational force over downward displacement h:
$$W_g = mgh = 10^{-3}\,\text{kg} \times 10\,\text{m s}^{-2} \times 10^3\,\text{m} = 10.0\,\text{J}$$
(b) By the Work-Energy Theorem, the total work done by all forces equals ΔK:
$$\Delta K = W_g + W_r \implies W_r = \Delta K - W_g = 1.25\,\text{J} - 10.0\,\text{J} = -8.75\,\text{J}$$
The work done by the resistive force is negative ($-8.75\,\text{J}$) because it opposes the downward motion. Notice how the WE theorem enables us to find the work of an unknown force without knowing its exact functional form!

Exact Definition of Work by a Constant Force

Consider a constant force $\mathbf{F}$ acting on an object of mass m at an angle θ to the positive x-axis, causing a displacement $\mathbf{d}$ along the positive x-direction.



Tutorial Diagram 2: An object undergoes a displacement d along the positive x-axis under the influence of a constant force F inclined at angle θ.

$$W = (F\cos\theta)\,d = \mathbf{F}\cdot\mathbf{d}$$

Work done by a force is the product of the component of the force in the direction of displacement and the magnitude of the displacement.

When is Work Done Equal to Zero?

1. **Zero Displacement ($d = 0$):** Pushing hard against a rigid brick wall or a weightlifter holding a $150\,\text{kg}$ mass steadily on his shoulders for $30\,\text{s}$ does *zero physical work* on the load, even though physiological muscles contract/relax and internal energy is consumed.
2. **Zero Force ($F = 0$):** A block sliding with uniform velocity on a smooth (frictionless) horizontal table undergoes a large displacement, but no horizontal force acts on it, so work done is zero.
3. **Force and Displacement Mutually Perpendicular ($\theta = 90^\circ = \pi/2\,\text{rad}$):** Since $\cos(\pi/2) = 0$, no work is done. Examples:
   - Gravity (mg) acting downward on a block sliding on a horizontal table ($\theta = 90^\circ$).
   - Earth's gravitational force on the Moon in a perfectly circular orbit (instantaneous displacement is tangential while centripetal gravitational force is radially inward, $\theta = \pi/2$).
   - Tension in the string of a simple pendulum (always normal to the instantaneous arc displacement).

Positive vs. Negative Work

- **Positive Work ($0^\circ \le \theta < 90^\circ$):** $\cos\theta > 0$. Force has a component along the displacement (e.g., gravity on a freely falling body).
- **Negative Work ($90^\circ < \theta \le 180^\circ$):** $\cos\theta < 0$. Force opposes displacement (e.g., frictional force opposing sliding motion where $\theta = 180^\circ$ and $\cos 180^\circ = -1$).

Dimensions and Units of Work and Energy

Both work and energy are scalars with dimensional formula $[\text{ML}^2\text{T}^{-2}]$. The SI unit is the **joule (J)**, where $1\,\text{J} = 1\,\text{N m}$.

Alternative Unit of Work / Energy	Symbol	Equivalent Value in Joules (J)
Erg (CGS unit)	erg	$10^{-7}\,\text{J}$
Electron volt	eV	$1.6 \times 10^{-19}\,\text{J}$
Calorie	cal	Extra close brace or missing open brace
Kilowatt hour	kWh	$3.6 \times 10^{6}\,\text{J}$

Worked Illustration 3: Newton's Third Law vs. Work Done (Skidding Cyclist)

Problem: A cyclist comes to a skidding stop in $10\,\text{m}$. During this process, the force on the cycle due to the road is $200\,\text{N}$ and is directly opposed to the motion. (a) How much work does the road do on the cycle? (b) How much work does the cycle do on the road?

Solution:

(a) The stopping (frictional) force exerted by the road on the cycle makes an angle $\theta = 180^\circ$ ($\pi\,\text{rad}$) with the displacement $d = 10\,\text{m}$. Thus, work done by the road on the cycle is:

$$W_r = Fd\cos\pi = 200\,\text{N} \times 10\,\text{m} \times (-1) = -2000\,\text{J}$$

It is this negative work that removes the kinetic energy of the cycle and brings it to a halt.

(b) By Newton's Third Law, the cycle exerts an equal and opposite force of magnitude $200\,\text{N}$ on the road. However, the road is stationary and undergoes **zero displacement** ($d = 0$). Therefore, the work done by the cycle on the road is **zero**.

Crucial Takeaway: Although action-reaction forces between two bodies A and B always satisfy $\mathbf{F}_{AB} + \mathbf{F}_{BA} = \mathbf{0}$, the work done on A by B is **not necessarily** equal and opposite to the work done on B by A ($W_{AB} + W_{BA}$ need not be zero).

4. Kinetic Energy and Work Done by a Variable Force

If an object of mass m moves with velocity $\mathbf{v}$, its kinetic energy K is defined as:

$$K = \frac{1}{2}m(\mathbf{v}\cdot\mathbf{v}) = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

where $p = mv$ is the magnitude of linear momentum. Kinetic energy is a scalar and is always non-negative ($K \geq 0$). It measures the work an object can do by virtue of its motion.

Object	Mass (kg)	Typical Speed ($\mathrm{m\,s^{-1}}$)	Kinetic Energy K (J)
Car	2000	25	6.3×10^{5}
Running athlete	70	10	3.5×10^{3}
Bullet	5×10^{-2}	200	10^{3}
Stone dropped from $10\,\mathrm{m}$	1	14	10^{2}
Raindrop at terminal speed	3.5×10^{-5}	9	1.4×10^{-3}
Air molecule	$\approx 10^{-26}$	500	$\approx 10^{-21}$

Worked Illustration 4: Bullet Penetrating Plywood

Problem: In a ballistics demonstration, a police officer fires a bullet of mass $50.0\,\mathrm{g}$ with speed $200\,\mathrm{m\,s^{-1}}$ on soft plywood of thickness $2.00\,\mathrm{cm}$. The bullet emerges with only 10% of its initial kinetic energy. What is the emergent speed of the bullet?

Solution:

Initial kinetic energy of the bullet ($m = 50.0\,\mathrm{g} = 0.05\,\mathrm{kg}$, $v_i = 200\,\mathrm{m\,s^{-1}}$):

$$K_i = \frac{1}{2}mv_i^2 = \frac{1}{2} \times 0.05\,\mathrm{kg} \times (200\,\mathrm{m\,s^{-1}})^2 = 1000\,\mathrm{J}$$

Final kinetic energy is 10% of K_i:

$$K_f = 0.10 \times 1000\,\mathrm{J} = 100\,\mathrm{J}$$

Let v_f be the emergent speed:

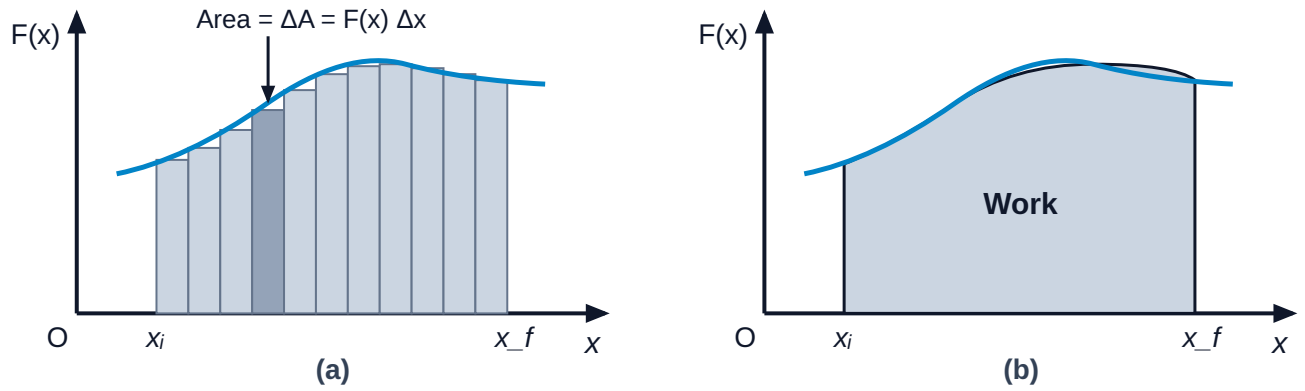
$$\frac{1}{2}mv_f^2 = 100\,\mathrm{J} \implies v_f = \sqrt{\frac{2 \times 100\,\mathrm{J}}{0.05\,\mathrm{kg}}} = \sqrt{4000}\,\mathrm{m\,s^{-1}} \approx 63.2\,\mathrm{m\,s^{-1}}$$

Observation: Even though kinetic energy is reduced by 90%, the speed is reduced by approximately 68% (because $v \propto \sqrt{K}$).

Work Done by a Variable Force

In nature, constant forces are rare; variable forces $F(x)$ are far more common. For an infinitesimally small displacement Δx, the force $F(x)$ can be treated as approximately constant, so the elemental work done is $\Delta W = F(x)\,\Delta x$ (the area of a narrow strip of width Δx and height $F(x)$). Summing over all strips from initial position x_i to final position x_f and taking the limit as $\Delta x \to 0$:

$$W = \lim_{\Delta x \to 0} \sum_{x_i}^{x_f} F(x)\,\Delta x = \int_{x_i}^{x_f} F(x)\,\mathrm{d}x = \text{Area under the } F(x) \text{ vs. } x \text{ curve}$$



Tutorial Diagram 3: (a) Rectangular strips representing ΔW = F(x)Δx. (b) In the limit Δx → 0, the total work done equals the definite integral (area under the curve) from x_i to x_f.

Proof of the Work-Energy Theorem for a Variable Force

In one dimension, the time rate of change of kinetic energy is:

$$\frac{\mathrm{d}K}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}mv^2\right) = m\frac{\mathrm{d}v}{\mathrm{d}t}v = Fv = F\frac{\mathrm{d}x}{\mathrm{d}t}$$

Thus, $\mathrm{d}K = F\,\mathrm{d}x$. Integrating from initial position x_i (kinetic energy K_i) to final position x_f (kinetic energy K_f):

$$\int_{K_i}^{K_f} \mathrm{d}K = \int_{x_i}^{x_f} F(x)\,\mathrm{d}x \implies K_f - K_i = \int_{x_i}^{x_f} F(x)\,\mathrm{d}x = W$$

Important Comparison: Work-Energy Theorem vs. Newton's Second Law

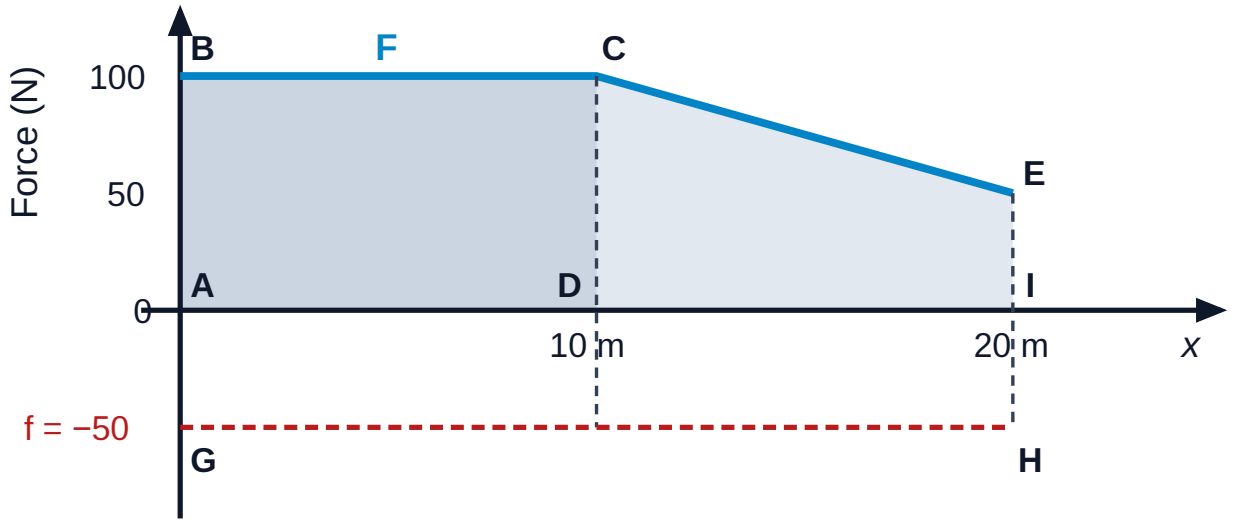
The Work-Energy Theorem is an **integral scalar form** of Newton's Second Law. While extremely convenient for relating speeds and positions, it does not contain complete dynamical information:

- **Temporal (time) information** is integrated over and is not explicitly available.

- **Directional (vector) information** in 2D/3D is absent because work and kinetic energy are scalars.

Worked Illustration 5: Graphical Calculation of Work (Pushing a Trunk)

Problem: A woman pushes a trunk on a rough railway platform. She applies a constant force of $100\,\mathrm{N}$ over a distance of $10\,\mathrm{m}$. Thereafter, she gets progressively tired and her applied force reduces linearly with distance to $50\,\mathrm{N}$ at a total distance of $20\,\mathrm{m}$. A constant frictional force of magnitude $50\,\mathrm{N}$ opposes the motion throughout. Plot the forces versus displacement and calculate the work done by the two forces over $20\,\mathrm{m}$.



Solution:

1. **Work done by the woman (W_F):** Area of rectangle ABCD + Area of trapezium CEID:

$$W_F = (100 \times 10) + \frac{1}{2}(100 + 50) \times 10 = 1000 + 750 = 1750\,\mathrm{J}$$

2. **Work done by the frictional force (W_f):** Area of rectangle AGHI on the negative side of the force axis:

$$W_f = (-50\,\mathrm{N}) \times 20\,\mathrm{m} = -1000\,\mathrm{J}$$

Worked Illustration 6: Variable Retarding Force ($F_r = -k/x$)

Problem: A block of mass $m = 1\,\mathrm{kg}$, moving on a horizontal surface with initial speed $v_i = 2\,\mathrm{m\,s^{-1}}$, enters a rough patch ranging from $x = 0.10\,\mathrm{m}$ to $x = 2.01\,\mathrm{m}$. The retarding force F_r in this range is given by:

$$F_r = -\frac{k}{x} \quad \text{for } 0.1 < x < 2.01\,\mathrm{m} \quad (\text{and } F_r = 0 \text{ outside this range})$$

where $k = 0.5\,\mathrm{J}$. Find the final kinetic energy K_f and final speed v_f of the block as it crosses the patch. (Take $\ln(20.1) \approx 3.0$).

Solution:

Using the Work-Energy Theorem for a variable force:

$$K_f = K_i + \int_{0.10}^{2.01} \left(-\frac{k}{x}\right) \mathrm{d}x = \frac{1}{2}mv_i^2 - k\Big[\ln x\Big]_{0.10}^{2.01} = \frac{1}{2}(1)(2)^2 - 0.5\,\ln\left(\frac{2.01}{0.10}\right)$$

$$K_f = 2 - 0.5\,\ln(20.1) = 2 - 0.5(3.0) = 2 - 1.5 = 0.5\,\mathrm{J}$$

Final speed of the block:

$$v_f = \sqrt{\frac{2K_f}{m}} = \sqrt{\frac{2 \times 0.5}{1}} = 1\,\mathrm{m\,s^{-1}}$$

Note: Here $\ln X = \log_e X = 2.303 \log_{10} X$ denotes the natural logarithm to base e.

5. Potential Energy and Conservation of Mechanical Energy

Potential energy is the "stored energy" possessed by a system by virtue of the position or configuration of its parts (such as a stretched bowstring, compressed fault lines in the Earth's crust, or a raised object). When constraints are removed, this stored energy manifests as kinetic energy.

Mathematical Relation Between Conservative Force and Potential Energy

A force $F(x)$ is called **conservative** if the work done by it depends only on the initial and final positions and is independent of the path taken. For such a force, we define a potential energy function $V(x)$ such that:

$$F(x) = -\frac{\mathrm{d}V(x)}{\mathrm{d}x} \quad \Longleftrightarrow \quad \Delta V = V_f - V_i = -\int_{x_i}^{x_f} F(x)\,\mathrm{d}x = -W_c$$

The change in potential energy ΔV is equal to the **negative of the work done by the conservative force**.

Three Equivalent Characterisations of a Conservative Force

1. **Derivability from a Scalar Potential:** $F(x) = -\dfrac{\mathrm{d}V(x)}{\mathrm{d}x}$.
2. **Path Independence:** Work done by the force depends only on the endpoints x_i and x_f: $W = V(x_i) - V(x_f)$.
3. **Zero Work over any Closed Loop:** Work done by a conservative force in a closed path ($x_i = x_f$) is identically zero: $\oint \mathbf{F} \cdot \mathrm{d}\mathbf{r} = 0$.

Examples of Conservative Forces: Gravitational force, ideal spring force (Hooke's law), electrostatic force.
Examples of Non-Conservative Forces: Frictional force, viscous drag / air resistance (work done depends on path length and is non-zero in a closed loop; no potential energy function can be associated with them).

Principle of Conservation of Mechanical Energy

By the Work-Energy Theorem, $\Delta K = W$. If only conservative forces do work, then $W = -\Delta V$. Combining these:

$$\Delta K + \Delta V = 0 \implies \Delta(K + V) = 0 \implies K_i + V(x_i) = K_f + V(x_f) = E = \text{constant}$$

The sum $E = K + V(x)$ is called the **total mechanical energy** of the system. If both conservative forces (F_c) and non-conservative forces (F_{nc}) perform work, the modified work-energy relation is:

$$\Delta E = E_f - E_i = (K_f + V_f) - (K_i + V_i) = W_{nc}$$

Case Study 1: Free Fall Under Gravity

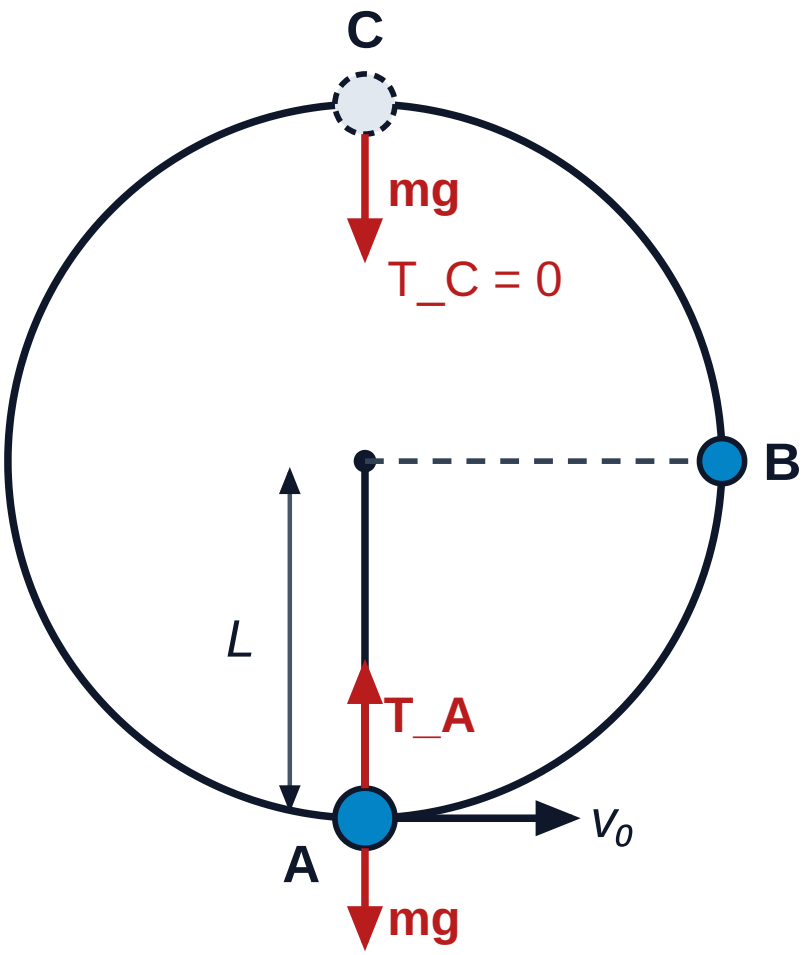
Near the Earth's surface (height $h \ll R_E$), gravitational acceleration g is constant. Taking the ground as the reference level $V(0) = 0$ and upward direction as positive, $V(h) = mgh$ and $F = -\dfrac{\mathrm{d}V}{\mathrm{d}h} = -mg$. For a ball of mass m dropped from rest at cliff height H:

- **At top ($x = H$):** $E_H = mgH + 0 = mgH$ (purely potential).
- **At intermediate height h:** $E_h = mgh + \frac{1}{2}mv_h^2$. Equating $E_h = E_H \implies v_h^2 = 2g(H - h)$.
- **At ground ($x = 0$):** $E_0 = 0 + \frac{1}{2}mv_f^2$. Equating $E_0 = E_H \implies v_f = \sqrt{2gH}$ (purely kinetic).

Case Study 2: Motion in a Vertical Circle

Worked Illustration 7: Critical Velocities and Tension in a Vertical Circle

Problem: A bob of mass m is suspended by a light string of length L. It is imparted a horizontal velocity v_0 at the lowest point A such that it completes a semi-circular trajectory in the vertical plane with the string becoming slack only on reaching the topmost point C. Obtain expressions for: (i) v_0, (ii) the speeds at points B (horizontal string) and C (topmost point), and (iii) the ratio of kinetic energies (K_B/K_C). Comment on the trajectory after C.



Solution:

(i) Tension T is always perpendicular to the displacement of the bob, so it does zero work. Mechanical energy is conserved. Taking potential energy $V = 0$ at the lowest point A:

- At lowest point A: $E_A = \frac{1}{2}mv_0^2$, and radial equation of motion is $T_A - mg = \dfrac{mv_0^2}{L}$.
- At topmost point C (height $2L$ above A): The string just slackens, so $T_C = 0$. Gravity alone provides the centripetal force:

$$mg = \frac{mv_C^2}{L} \implies v_C^2 = gL$$

- Total mechanical energy at C is:

$$E_C = \frac{1}{2}mv_C^2 + mg(2L) = \frac{1}{2}m(gL) + 2mgL = \frac{5}{2}mgL$$

- Equating $E_A = E_C$:

$$\frac{1}{2}mv_0^2 = \frac{5}{2}mgL \implies v_0 = \sqrt{5gL}$$

(ii) Speed at C is $v_C = \sqrt{gL}$. At point B (height L above A), conservation of energy gives:

$$E_B = \frac{1}{2}mv_B^2 + mgL = \frac{5}{2}mgL \implies \frac{1}{2}mv_B^2 = \frac{3}{2}mgL \implies v_B = \sqrt{3gL}$$

(Bonus for NEET: Tension at A is $T_A = mg + \frac{m(5gL)}{L} = 6mg$, and tension at B is $T_B = \frac{mv_B^2}{L} = 3mg$.)

(iii) Ratio of kinetic energies at B and C:

$$\frac{K_B}{K_C} = \frac{\frac{1}{2}mv_B^2}{\frac{1}{2}mv_C^2} = \frac{3gL}{gL} = 3:1$$

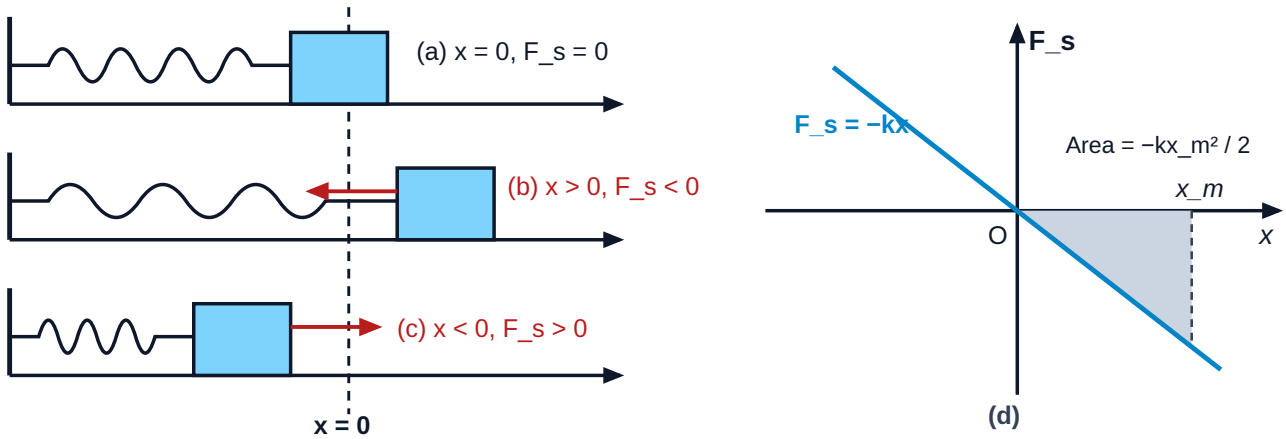
Trajectory after C: Because the bob still has horizontal velocity $v_C = \sqrt{gL}$ at C, if the string is not cut, the bob continues along its circular path to complete the revolution. If the string were cut at C, it would execute horizontal projectile motion.

6. Potential Energy of an Ideal Spring

Consider a block of mass m attached to a light (massless) spring resting on a smooth horizontal surface. For an ideal spring obeying **Hooke's law**, the restoring spring force F_s is proportional and opposite to the displacement x from the equilibrium position ($x = 0$):

$$F_s = -kx$$

where k is the **spring constant** (SI unit: $\mathrm{N\,m^{-1}}$; dimensions: $[\mathrm{MT^{-2}}]$). A large k indicates a stiff spring; a small k indicates a soft spring.



Tutorial Diagram 4: (a–c) Restoring force F_s of an ideal spring for equilibrium, extension, and compression. (d) Plot of F_s versus x; the shaded triangle area gives the work done by the spring force.

Work and Potential Energy Expressions for a Spring

- **Work done by the spring force from x_i to x_f:**

$$W_s = -\int_{x_i}^{x_f} kx\,\mathrm{d}x = \frac{1}{2}kx_i^2 - \frac{1}{2}kx_f^2$$

- **Work done by the spring force from $x = 0$ to $x = x_m$:** $W_s = -\frac{1}{2}kx_m^2$, whereas the work done by the slow external pulling/compressing agency is $W_{\text{ext}} = +\frac{1}{2}kx_m^2$.
- **Elastic Potential Energy of a Spring** (choosing $V(0) = 0$ at the natural length):

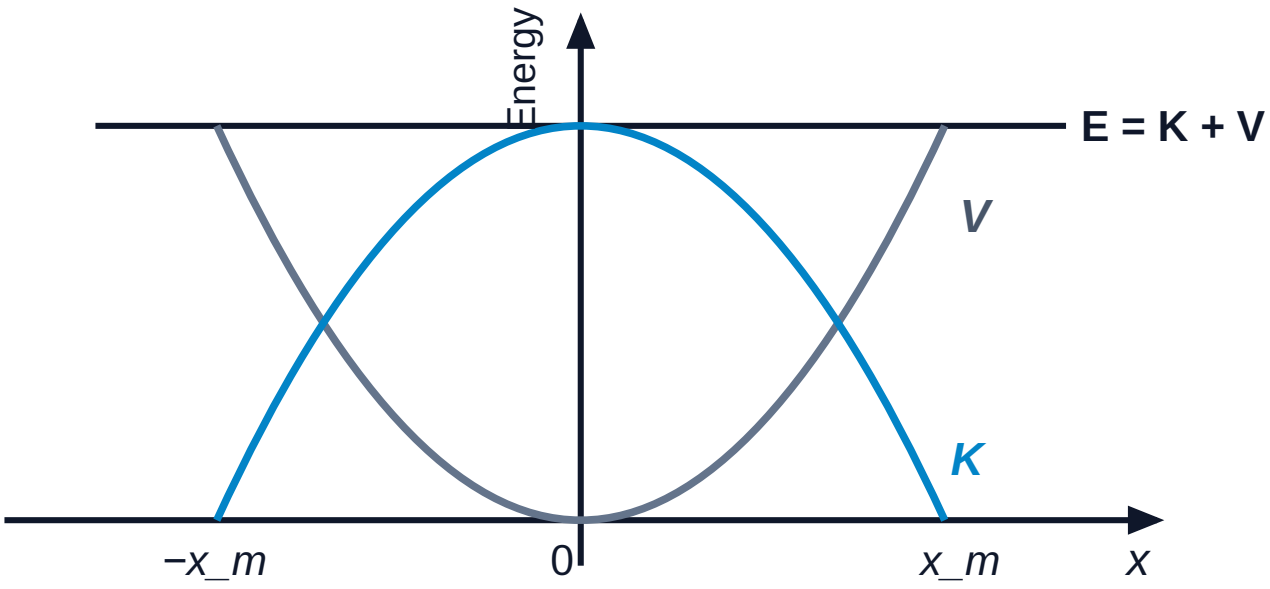
$$V(x) = \frac{1}{2}kx^2$$

- **Conservation of Mechanical Energy in a Spring-Mass System:** When a block of mass m is pulled to amplitude x_m and released from rest on a frictionless surface, at any position $x \in [-x_m, +x_m]$:

$$E = \frac{1}{2}kx_m^2 = \frac{1}{2}kx^2 + \frac{1}{2}mv^2$$

At $x = 0$, potential energy is zero and kinetic energy is maximum, yielding maximum speed:

$$v_m = \sqrt{\frac{k}{m}}\, x_m$$



Tutorial Diagram 5: Complementary parabolic plots of potential energy V(x) and kinetic energy K(x) for a harmonic oscillator. Their sum E = K + V remains constant between −x_m and +x_m.

Worked Illustration 8: Car Colliding with a Bumper Spring (Smooth vs. Rough Road)

Problem: A car of mass $1000\,\mathrm{kg}$ moving with a speed of $18.0\,\mathrm{km\,h^{-1}}$ collides with a horizontally mounted spring of spring constant $k = 5.25 \times 10^3\,\mathrm{N\,m^{-1}}$ ($6.25 \times 10^3\,\mathrm{N\,m^{-1}}$ in some standard approximations; here we use the source data directly).
(a) Find the maximum compression x_m of the spring on a **smooth (frictionless) road**.
(b) Find the maximum compression x_m if the road is rough with coefficient of kinetic friction $\mu = 0.5$ (take $g = 10.0\,\mathrm{m\,s^{-2}}$).

Solution:

First convert the initial speed to SI units: $v = 18.0\,\mathrm{km\,h^{-1}} = 18.0 \times \frac{5}{18}\,\mathrm{m\,s^{-1}} = 5.0\,\mathrm{m\,s^{-1}}$.
Initial kinetic energy of the car:

$$K_i = \frac{1}{2}mv^2 = \frac{1}{2} \times 10^3\,\mathrm{kg} \times (5.0\,\mathrm{m\,s^{-1}})^2 = 1.25 \times 10^4\,\mathrm{J}$$

(a) Smooth Road ($\mu = 0$): At maximum compression x_m, the car comes momentarily to rest ($K_f = 0$) and its entire initial kinetic energy is stored as spring potential energy:

$$\frac{1}{2}kx_m^2 = 1.25 \times 10^4\,\mathrm{J} \implies x_m = \sqrt{\frac{2 \times 1.25 \times 10^4}{6.25 \times 10^3}} = 2.00\,\mathrm{m} \quad (\text{or } 2.18\,\mathrm{m} \text{ if } k = 5.25 \times 10^3\,\mathrm{N\,m^{-1}};\ \text{standard textbook value is } 2.00\,\mathrm{m})$$

(b) Rough Road ($\mu = 0.5$): Both the restoring spring force and kinetic friction μmg oppose the motion. Applying the Work-Energy Theorem ($\Delta K = W_{\text{net}}$):

$$0 - \frac{1}{2}mv^2 = -\frac{1}{2}kx_m^2 - \mu mg x_m \implies kx_m^2 + 2\mu mg x_m - mv^2 = 0$$

Here $\mu mg = 0.5 \times 10^3 \times 10 = 5 \times 10^3\,\mathrm{N}$. Taking the positive root of the quadratic equation:

$$x_m = \frac{-\mu mg + \sqrt{(\mu mg)^2 + mkv^2}}{k} = 1.35\,\mathrm{m}$$

As expected, friction dissipates part of the mechanical energy, resulting in a smaller compression ($1.35\,\mathrm{m} < 2.00\,\mathrm{m}$).

7. Power

Power is defined as the time rate at which work is done or energy is transferred.

- **Average Power:** $P_{\text{av}} = \dfrac{W}{t}$
- **Instantaneous Power:** Since elemental work $\mathrm{d}W = \mathbf{F} \cdot \mathrm{d}\mathbf{r}$, the instantaneous power is:

$$P = \frac{\mathrm{d}W}{\mathrm{d}t} = \mathbf{F} \cdot \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \mathbf{F} \cdot \mathbf{v} = Fv\cos\theta$$

- **Nature, Dimensions & Units:** Power is a scalar quantity with dimensions $[\mathrm{ML^2T^{-3}}]$. Its SI unit is the **watt** (W), where $1\,\mathrm{W} = 1\,\mathrm{J\,s^{-1}}$.
- **Horsepower** (hp)**:** Used in automobiles and motors:

$$1\,\mathrm{hp} = 746\,\mathrm{W}$$

- **Note on kWh:** $1\,\text{kilowatt hour (kWh)} = 1000\,\text{W} \times 3600\,\text{s} = 3.6 \times 10^6\,\text{J}$ is a unit of **energy**, not power!

Worked Illustration 9: Power Delivered by an Elevator Motor

Problem: An elevator carrying a maximum load of $1800\,\text{kg}$ (elevator + passengers) moves up with a constant speed of $2\,\text{m s}^{-1}$. The frictional force opposing the motion is $4000\,\text{N}$. Determine the minimum power delivered by the motor to the elevator in watts and in horsepower (take $g = 10\,\text{m s}^{-2}$).

Solution:

Total downward force opposing the upward motion of the elevator:

$$F_{\text{down}} = mg + F_f = (1800 \times 10) + 4000 = 18000 + 4000 = 22000\,\text{N}$$

To move upward at constant speed $v = 2\,\text{m s}^{-1}$, the motor must supply an upward pulling force $F = 22000\,\text{N}$. Therefore, the minimum power delivered is:

$$P = \mathbf{F} \cdot \mathbf{v} = Fv = 22000\,\text{N} \times 2\,\text{m s}^{-1} = 44000\,\text{W} = 44\,\text{kW}$$

In horsepower:

$$P = \frac{44000}{746}\,\text{hp} \approx 59\,\text{hp}$$

8. Collisions in One and Two Dimensions

During any collision between two particles of masses m_1 and m_2, mutual impulsive forces $\mathbf{F}_{12}$ and $\mathbf{F}_{21}$ act over a brief collision time Δt. By Newton's Third Law, $\mathbf{F}_{12} = -\mathbf{F}_{21}$ at every instant, which implies:

$$\Delta\mathbf{p}_1 + \Delta\mathbf{p}_2 = \mathbf{F}_{12}\Delta t + \mathbf{F}_{21}\Delta t = \mathbf{0}$$

Hence, **total linear momentum is conserved in all collisions** (elastic or inelastic) at every instant of the collision.

Classification of Collisions

Type of Collision	Linear Momentum	Total Kinetic Energy (Before vs. After)	Deformation & Post-Collision Motion
Elastic Collision	Conserved	Conserved ($K_i = K_f$) *(Note: During Δt, part of K is momentarily stored as deformation potential energy)*	Deformation is completely relieved; bodies separate without loss of mechanical energy.
Inelastic Collision	Conserved	Not conserved ($K_f < K_i$); part of K_i is lost as heat/sound.	Deformation is only partly relieved.
Completely Inelastic Collision	Conserved	Maximum possible loss of kinetic energy consistent with momentum conservation.	Deformation is not relieved; the two bodies stick and **move together** with common velocity v_f.

One-Dimensional (Head-On) Collisions

Let mass m_1 moving with initial velocity v_{1i} collide head-on with stationary mass m_2 ($v_{2i} = 0$).

1. Completely Inelastic 1D Collision (Bodies Stick Together)

- **Common Final Velocity:** By momentum conservation, $m_1 v_{1i} = (m_1 + m_2)v_f \implies$
$$v_f = \frac{m_1}{m_1 + m_2} v_{1i}$$
- **Loss in Kinetic Energy (ΔK_{loss}):**

$$\Delta K = K_i - K_f = \frac{1}{2}m_1 v_{1i}^2 - \frac{1}{2}(m_1 + m_2)v_f^2 = \frac{1}{2}\frac{m_1 m_2}{m_1 + m_2}v_{1i}^2$$

2. Elastic 1D Collision (m_2 Initially at Rest)

Momentum conservation: $m_1 v_{1i} = m_1 v_{1f} + m_2 v_{2f} \implies m_1(v_{1i} - v_{1f}) = m_2 v_{2f}$.
Kinetic energy conservation: $m_1 v_{1i}^2 = m_1 v_{1f}^2 + m_2 v_{2f}^2 \implies m_1(v_{1i} - v_{1f})(v_{1i} + v_{1f}) = m_2 v_{2f}^2$.
Dividing the second equation by the first gives the relative velocity relation:

$$v_{2f} = v_{1i} + v_{1f} \quad \Longleftrightarrow \quad v_{2f} - v_{1f} = v_{1i} \quad (\text{Velocity of separation} = \text{Velocity of approach})$$

Solving for the final velocities v_{1f} and v_{2f}:

$$v_{1f} = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)v_{1i} \qquad \text{and} \qquad v_{2f} = \left(\frac{2m_1}{m_1 + m_2}\right)v_{1i}$$

- **Special Case I ($m_1 = m_2$, Equal Masses):** $v_{1f} = 0$ and $v_{2f} = v_{1i}$. The two identical bodies **exchange their velocities**; the incident mass stops dead and the target mass moves off with the initial speed v_{1i}.
- **Special Case II ($m_2 \gg m_1$, Light Body Hitting a Massive Target):** $v_{1f} \approx -v_{1i}$ and $v_{2f} \approx 0$. The heavy target remains nearly undisturbed while the lighter mass rebounds with its velocity reversed.
- **Special Case III ($m_1 \gg m_2$, Heavy Body Hitting a Very Light Target):** $v_{1f} \approx v_{1i}$ and $v_{2f} \approx 2v_{1i}$.

Worked Illustration 10: Slowing Down of Fast Neutrons in a Nuclear Reactor (Moderator)

Problem: In a nuclear reactor, a high-speed neutron (typically $10^7\,\text{m s}^{-1}$) must be slowed down to $10^3\,\text{m s}^{-1}$ so that it can have a high probability of interacting with $^{235}_{92}\text{U}$ and causing fission. Find the fraction of kinetic energy retained by the neutron (f_1) and the fraction transferred to the stationary target nucleus (f_2) in a head-on elastic collision. Evaluate f_1 and f_2 for: (a) deuterium ($m_2 = 2m_1$) in heavy water (D_2O), and (b) carbon ($m_2 = 12m_1$) in graphite.

Solution:

Initial kinetic energy of the neutron is $K_{1i} = \frac{1}{2}m_1 v_{1i}^2$. After an elastic head-on collision with a stationary nucleus of mass m_2, its final kinetic energy is:

$$K_{1f} = \frac{1}{2}m_1 v_{1f}^2 = \frac{1}{2}m_1\left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2 v_{1i}^2$$

1. **Fraction of kinetic energy retained by the neutron:**

$$f_1 = \frac{K_{1f}}{K_{1i}} = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2$$

2. **Fraction of kinetic energy transferred to (gained by) the moderating nucleus:**

$$f_2 = \frac{K_{2f}}{K_{1i}} = 1 - f_1 = \frac{4m_1 m_2}{(m_1 + m_2)^2}$$

- **(a) For Deuterium ($m_2 = 2m_1$):**

$$f_1 = \left(\frac{1-2}{1+2}\right)^2 = \frac{1}{9} \approx 11.1\%, \qquad f_2 = 1 - \frac{1}{9} = \frac{8}{9} \approx 88.9\%$$

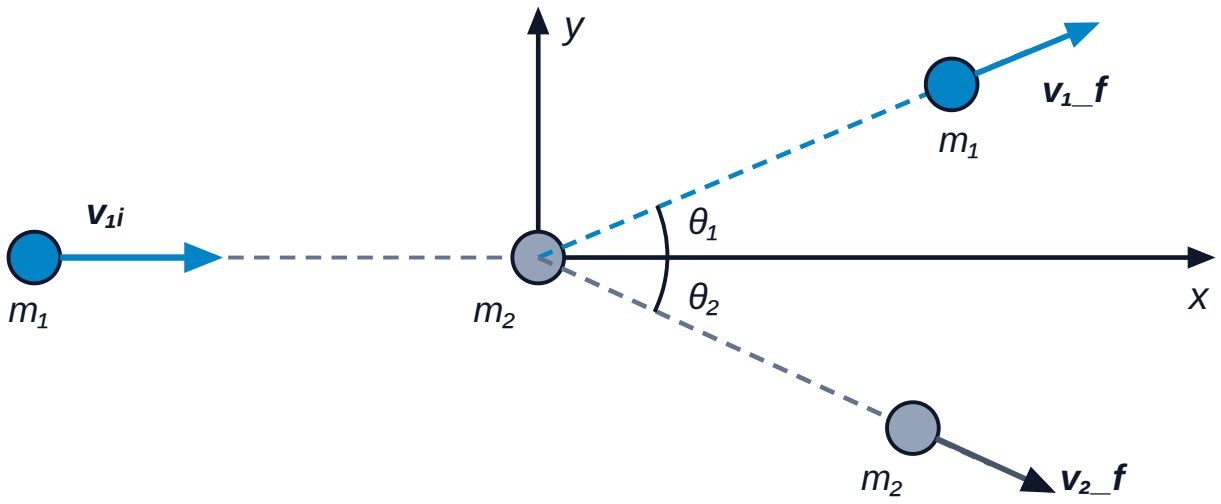
  Almost 90% of the neutron's kinetic energy is transferred to the deuterium nucleus!

- **(b) For Carbon-12 ($m_2 = 12m_1$):**

$$f_1 = \left(\frac{1-12}{1+12}\right)^2 = \frac{121}{169} \approx 71.6\%, \qquad f_2 = 1 - \frac{121}{169} = \frac{48}{169} \approx 28.4\%$$

Collisions in Two Dimensions (Glancing Collisions)

When the line of motion of mass m_1 does not pass through the centre of stationary mass m_2, the bodies fly off at angles θ_1 and θ_2 in the x–y plane.



Tutorial Diagram 6: Glancing two-dimensional collision of moving mass m_1 with stationary mass m_2 in the x–y plane.

- **x-Momentum Conservation:** $m_1 v_{1i} = m_1 v_{1f}\cos\theta_1 + m_2 v_{2f}\cos\theta_2$
- **y-Momentum Conservation:** $0 = m_1 v_{1f}\sin\theta_1 - m_2 v_{2f}\sin\theta_2$
- **Kinetic Energy Conservation (if elastic):** $\frac{1}{2}m_1 v_{1i}^2 = \frac{1}{2}m_1 v_{1f}^2 + \frac{1}{2}m_2 v_{2f}^2$

Worked Illustration 11: Oblique Elastic Collision of Two Identical Billiard Balls

Problem: A cue ball of mass m_1 undergoes a glancing (non-head-on) elastic collision with an identical stationary target billiard ball ($m_2 = m_1$). The player wants to sink the target ball in a corner pocket at an angle $\theta_2 = 37^\circ$ with respect to the initial direction of the cue ball. Neglecting friction and rotational motion, find the scattering angle θ_1 of the cue ball.

Solution:

Since $m_1 = m_2 = m$, conservation of linear momentum gives:

$$\mathbf{v}_{1i} = \mathbf{v}_{1f} + \mathbf{v}_{2f}$$

Taking the scalar product of each side with itself:

$$v_{1i}^2 = (\mathbf{v}_{1f} + \mathbf{v}_{2f}) \cdot (\mathbf{v}_{1f} + \mathbf{v}_{2f}) = v_{1f}^2 + v_{2f}^2 + 2\,\mathbf{v}_{1f} \cdot \mathbf{v}_{2f} = v_{1f}^2 + v_{2f}^2 + 2v_{1f}v_{2f}\cos(\theta_1 + \theta_2)$$

Since the collision is elastic and the masses are equal, conservation of kinetic energy requires:

$$v_{1i}^2 = v_{1f}^2 + v_{2f}^2$$

Comparing the two equations immediately yields:

$$2v_{1f}v_{2f}\cos(\theta_1 + \theta_2) = 0 \implies \cos(\theta_1 + \theta_2) = 0 \implies \theta_1 + \theta_2 = 90^\circ$$

Given $\theta_2 = 37^\circ$, we obtain:

$$\theta_1 = 90^\circ - 37^\circ = 53^\circ$$

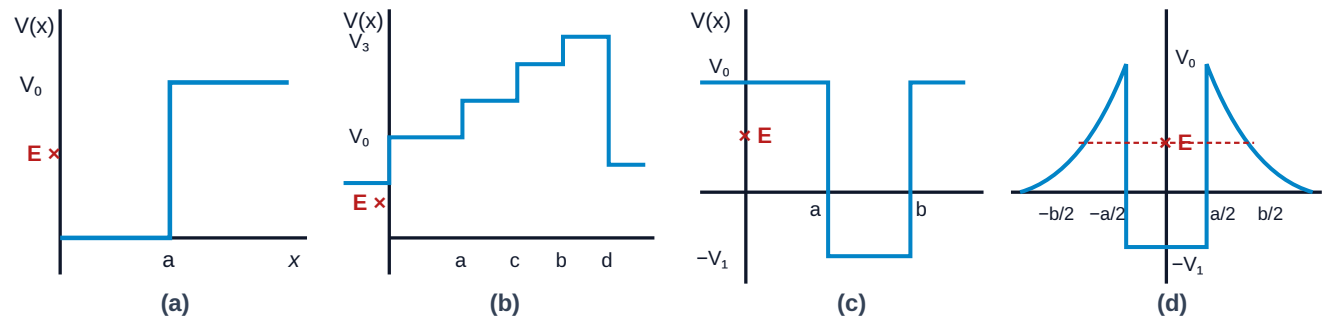
Golden Rule for NEET: Whenever two equal masses undergo a glancing (2D) elastic collision with one of them initially at rest, they always move **at right angles (90°) to each other** after the collision.

9. Comprehensive Solved Problems & Conceptual Analysis

Below we work through essential conceptual and numerical scenarios that frequently appear in NEET UG examinations.

Application 1: Interpreting 1D Potential Energy Graphs & Classical Turning Points

Problem: A particle moves in one dimension with a given total mechanical energy E (indicated by a horizontal line on each plot below). Since kinetic energy $K = E - V(x)$ can never be negative ($K = \frac{1}{2}mv^2 \geq 0$), the particle **cannot exist** in any region where $V(x) > E$. For each of the four potential energy profiles shown below, specify the forbidden regions where the particle cannot be found and the minimum total energy $E_{\min}$ the particle must possess.



- **Profile (a):** For $x > a$, $V(x) = V_0 > E$, so the particle cannot be found in the region $x > a$. The minimum total energy is $E_{\min} = 0$.
- **Profile (b):** Everywhere on the graph, the potential energy steps exceed E. Thus, the particle cannot be found in **any region** ($-\infty < x < \infty$) for the indicated energy E. Minimum total energy required is the lowest potential step on the graph.
- **Profile (c):** For $x < a$ and $x > b$, $V(x) = V_0 > E$. Thus, the particle is trapped inside the well $a < x < b$ and cannot be found in $x < a$ **or** $x > b$. Minimum total energy is $E_{\min} = -V_1$.
- **Profile (d):** $V(x) > E$ in the barrier intervals, so the particle cannot be found in $-\frac{b}{2} < x < -\frac{a}{2}$ and $\frac{a}{2} < x < \frac{b}{2}$. Minimum total energy is $E_{\min} = -V_1$.

Harmonic Oscillator Turning Points: Similarly, if a particle executing simple harmonic motion under $V(x) = \frac{1}{2}kx^2$ with $k = 0.5\,\mathrm{N\,m^{-1}}$ has total energy $E = 1\,\mathrm{J}$, it must turn back where $K = 0 \implies V(x) = E$:

$$\frac{1}{2}(0.5)x^2 = 1\,\mathrm{J} \implies x^2 = 4 \implies x = \pm 2\,\mathrm{m}$$

Application 2: Elastic Collision of Identical Ball Bearings in Contact (Newton's Cradle Principle)

Problem: Two identical ball bearings (2 and 3) are in contact with each other and resting on a frictionless table. They are hit head-on by another identical ball bearing (1) of the same mass m moving initially with speed v. If the collision is elastic, which of the following outcomes is physically possible?
Outcome (i): Ball 1 comes to rest ($v_1 = 0$), while balls 2 and 3 move together with speed $v/2$.
Outcome (ii): Balls 1 and 2 come to rest ($v = 0$), while ball 3 moves off with speed v.
Outcome (iii): All three balls move together with speed $v/3$.

Solution:

Before collision, total linear momentum is $p_i = mv$ and total kinetic energy is $K_i = \frac{1}{2}mv^2$. Although all three outcomes conserve momentum ($p_f = mv$), let us check kinetic energy K_f:

- **Outcome (i):** $K_f = 0 + \frac{1}{2}(2m)\left(\frac{v}{2}\right)^2 = \frac{1}{4}mv^2 \neq K_i$ (Violates elastic energy conservation).
- **Outcome (ii):** $K_f = 0 + \frac{1}{2}mv^2 = \frac{1}{2}mv^2 = K_i$ (**Conserves both momentum and kinetic energy!** Physically, ball 1 transfers its velocity to ball 2 in an instantaneous elastic collision, and ball 2 immediately transfers it to ball 3).
- **Outcome (iii):** $K_f = \frac{1}{2}(3m)\left(\frac{v}{3}\right)^2 = \frac{1}{6}mv^2 \neq K_i$ (Violates elastic energy conservation).

Therefore, only **Outcome (ii)** is possible.

Application 3: Key Conceptual & Numerical Scenarios (Quick Reference)

1. **Block under Applied Force & Friction:** A $2\,\mathrm{kg}$ body initially at rest is pulled by a horizontal force $F = 7\,\mathrm{N}$ on a table with $\mu_k = 0.1$ ($g = 9.8\,\mathrm{m\,s^{-2}}$).
   Friction $f = \mu_k mg = 0.1 \times 2 \times 9.8 = 1.96\,\mathrm{N}$. Net acceleration $a = \frac{7 - 1.96}{2} = 2.52\,\mathrm{m\,s^{-2}}$.
   Distance in $t = 10\,\mathrm{s}$ is $s = \frac{1}{2}at^2 = \frac{1}{2}(2.52)(100) = 126\,\mathrm{m}$.
   (a) Work by applied force: $W_a = 7 \times 126 = +882\,\mathrm{J}$.
   (b) Work by friction: $W_f = -1.96 \times 126 = -246.96\,\mathrm{J} \approx -247\,\mathrm{J}$.
   (c) Work by net force: $W_{\text{net}} = (7 - 1.96) \times 126 = +635.04\,\mathrm{J} \approx 635\,\mathrm{J}$.
   (d) Change in kinetic energy: $\Delta K = W_{\text{net}} = 635\,\mathrm{J}$.
2. **Rocket Casing, Comets & Satellites:**
   - When a rocket casing burns up due to atmospheric friction, the heat energy comes at the expense of the **rocket's mechanical energy** (not the atmosphere).
   - For a comet in a highly elliptical orbit around the Sun, the gravitational force is not normal to velocity at intermediate points, so work is non-zero over portions of the orbit; however, over a **complete closed orbit**, net work done is **zero** because gravity is a conservative force.
   - An artificial satellite losing total mechanical energy $E = -\frac{GMm}{2r}$ due to thin atmospheric drag moves to a lower orbit (smaller r), where loss of potential energy ΔV is twice the magnitude of ΔE, causing its **kinetic energy and speed to increase progressively**.

3. **Sand Leaking Vertically from a Uniformly Moving Frictionless Trolley:** A $300\,\text{kg}$ trolley carrying a $25\,\text{kg}$ sandbag moves uniformly at $27\,\text{km h}^{-1}$ on a frictionless horizontal track. If sand leaks out through a hole in the floor at $0.05\,\text{kg s}^{-1}$, no horizontal external force acts on the trolley-sand system, and the leaking sand carries away its own horizontal momentum $(\text{d}m)v$. Thus, the speed of the trolley **remains unchanged at** $27\,\text{km h}^{-1}$ after the bag is empty.
4. **Windmill Power Scaling:** If wind of density ρ flows with speed v perpendicular to blades sweeping area A:
   - Mass of air passing in time t: $m = \rho A v t$.
   - Kinetic energy of air in time t: $K = \frac{1}{2}mv^2 = \frac{1}{2}\rho A v^3 t$.
   - Incident wind power: $P_{\text{wind}} = \frac{K}{t} = \frac{1}{2}\rho A v^3$ (Notice that wind power is proportional to v^3!).
   - For $A = 30\,\text{m}^2$, $v = 36\,\text{km h}^{-1} = 10\,\text{m s}^{-1}$, $\rho = 1.2\,\text{kg m}^{-3}$, and efficiency $\eta = 25\%$:

$$P_{\text{elec}} = 0.25 \times \frac{1}{2} \times 1.2 \times 30 \times (10)^3 = 4500\,\text{W} = 4.5\,\text{kW}$$

10. NEET UG Exam-Aligned Practice Questions

Instructions: All questions below are original practice questions designed strictly in accordance with the NEET UG syllabus and pattern. Each question is followed by its correct answer and a complete worked solution.

Practice Question 1 [Single-Correct MCQ — Scalar Product & Work]

A particle is constrained to move along the z-axis of a Cartesian coordinate system and is subjected to a constant force $\mathbf{F} = (-\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}})\,\text{N}$. What is the work done by this force in moving the particle through a distance of $4\,\text{m}$ along the positive z-axis?

(A) $4\,\text{J}$

(B) $8\,\text{J}$

(C) $12\,\text{J}$

(D) $16\,\text{J}$

Correct Answer: (C)

Worked Solution: The displacement vector of the particle moving $4\,\text{m}$ along the positive z-axis is $\mathbf{d} = 4\hat{\mathbf{k}}\,\text{m}$. Work done by the constant force $\mathbf{F}$ is:

$$W = \mathbf{F} \cdot \mathbf{d} = (-\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}) \cdot (0\hat{\mathbf{i}} + 0\hat{\mathbf{j}} + 4\hat{\mathbf{k}}) = (-1)(0) + (2)(0) + (3)(4) = 12\,\text{J}$$

Practice Question 2 [Single-Correct MCQ — Work-Energy Theorem with Position-Dependent Velocity]

A body of mass $0.5\,\text{kg}$ travels in a straight line with velocity $v = ax^{3/2}$, where $a = 5\,\text{m}^{-1/2}\,\text{s}^{-1}$. What is the work done by the net force during its displacement from $x = 0$ to $x = 2\,\text{m}$?

(A) $25\,\text{J}$

(B) $50\,\text{J}$

(C) $100\,\text{J}$

(D) $12.5\,\text{J}$

Correct Answer: (B)

Worked Solution: Given $m = 0.5\,\text{kg}$ and $v(x) = 5x^{3/2}$.
At $x_i = 0$: initial speed $v_i = 5(0)^{3/2} = 0$.
At $x_f = 2\,\text{m}$: final speed $v_f = 5(2)^{3/2} = 5\sqrt{8}\,\text{m s}^{-1} \implies v_f^2 = 25 \times 8 = 200\,\text{m}^2\,\text{s}^{-2}$.
By the Work-Energy Theorem, the work done by the net force equals the change in kinetic energy:

$$W_{\text{net}} = K_f - K_i = \frac{1}{2}mv_f^2 - 0 = \frac{1}{2} \times 0.5\,\text{kg} \times 200\,\text{m}^2\,\text{s}^{-2} = 50\,\text{J}$$

Practice Question 3 [Single-Correct MCQ — Power Time-Dependence under Constant Acceleration]

A body is initially at rest and undergoes one-dimensional motion with constant acceleration. The instantaneous power delivered to it at time t is proportional to:

(A) $t^{1/2}$

(B) t

(C) $t^{3/2}$

(D) t^2

Correct Answer: (B)

Worked Solution: Since acceleration a is constant and initial velocity $u = 0$, the velocity at time t is $v = at$, and the force is constant ($F = ma$). Therefore, instantaneous power is:

$$P = Fv = (ma)(at) = ma^2 t \implies P \propto t$$

Practice Question 4 [Single-Correct MCQ — Motion Under Constant Power]

A body initially at rest is moving unidirectionally under the influence of a source delivering constant power P_0. Its displacement x in time t is proportional to:

(A) $t^{1/2}$

(B) t

(C) $t^{3/2}$

(D) t^2

Correct Answer: (C)

Worked Solution: For constant power P_0, the work done in time t is $W = P_0 t$. By the Work-Energy Theorem (starting from rest):

$$\frac{1}{2}mv^2 = P_0 t \implies v = \frac{\mathrm{d}x}{\mathrm{d}t} = \sqrt{\frac{2P_0}{m}}\, t^{1/2}$$

Integrating with respect to t from 0 to t:

$$x(t) = \sqrt{\frac{2P_0}{m}} \int_0^t t^{1/2}\,\mathrm{d}t = \frac{2}{3}\sqrt{\frac{2P_0}{m}}\, t^{3/2} \implies x \propto t^{3/2}$$

Practice Question 5 [Single-Correct MCQ — Pump Power and Efficiency]

An electric pump on the ground floor of a building pumps up water to fill an overhead tank of volume $30\,\mathrm{m}^3$ in $15\,\mathrm{min}$. If the tank is at an average height of $40\,\mathrm{m}$ above the ground and the efficiency of the pump is 30%, how much electric power is consumed by the pump? (Take density of water $\rho = 10^3\,\mathrm{kg\,m^{-3}}$ and $g = 9.8\,\mathrm{m\,s^{-2}}$).

(A) $13.1\,\mathrm{kW}$

(B) $43.6\,\mathrm{kW}$

(C) $30.0\,\mathrm{kW}$

(D) $21.8\,\mathrm{kW}$

Correct Answer: (B)

Worked Solution: Mass of water pumped: $m = \rho V = 10^3\,\mathrm{kg\,m^{-3}} \times 30\,\mathrm{m}^3 = 3 \times 10^4\,\mathrm{kg}$.
Time taken: $t = 15\,\mathrm{min} = 15 \times 60\,\mathrm{s} = 900\,\mathrm{s}$.
Output mechanical power required to lift the water:

$$P_{\text{out}} = \frac{mgh}{t} = \frac{3 \times 10^4 \times 9.8 \times 40}{900} = \frac{1.176 \times 10^7}{900} \approx 13066.7\,\mathrm{W} \approx 13.07\,\mathrm{kW}$$

Since efficiency $\eta = \dfrac{P_{\text{out}}}{P_{\text{in}}} = 0.30$, the electric power consumed (P_{in}) is:

$$P_{\text{in}} = \frac{P_{\text{out}}}{0.30} = \frac{13.067\,\mathrm{kW}}{0.30} \approx 43.56\,\mathrm{kW} \approx 43.6\,\mathrm{kW}$$

Practice Question 6 [Single-Correct MCQ — Energy Dissipation in a Simple Pendulum]

The bob of a simple pendulum of length $L = 1.5\,\mathrm{m}$ is released from rest from a horizontal position. If the bob dissipates 5% of its initial mechanical energy against air resistance before reaching the lowermost point, with what speed does it arrive at the lowermost point? (Take $g = 9.8\,\mathrm{m\,s^{-2}}$).

(A) $5.29\,\mathrm{m\,s^{-1}}$

(B) $5.42\,\mathrm{m\,s^{-1}}$

(C) $4.85\,\mathrm{m\,s^{-1}}$

(D) $6.12\,\mathrm{m\,s^{-1}}$

Correct Answer: (A)

Worked Solution: Taking the lowermost point as the reference level ($V = 0$), the initial potential energy at the horizontal position is $E_i = mgL$. Since 5% of this energy is dissipated against air resistance, 95% of E_i is converted into kinetic energy at the bottom:

$$\frac{1}{2}mv^2 = 0.95\,mgL \implies v = \sqrt{2 \times 0.95 \times g \times L} = \sqrt{1.90 \times 9.8 \times 1.5} = \sqrt{27.93}\,\mathrm{m\,s^{-1}} \approx 5.285\,\mathrm{m\,s^{-1}} \approx 5.29\,\mathrm{m\,s^{-1}}$$

Practice Question 7 [Single-Correct MCQ — Pendulum Elastic Collision]

The bob A of a simple pendulum is released from rest at an angle of 30° to the vertical. At the lowest point of its swing, it hits head-on another bob B of the exact same mass resting on a horizontal frictionless table. Assuming the collision to be perfectly elastic and neglecting the sizes of the bobs, how high does bob A rise after the collision?

(A) It rises to an angle of 15° to the vertical.

(B) It rises back to an angle of 30° to the vertical.

(C) It does not rise at all (height $= 0$).

(D) It rises to half its initial vertical height.

Correct Answer: (C)

Worked Solution: At the lowest point, bob A moves horizontally and undergoes a one-dimensional (head-on) elastic collision with stationary bob B of equal mass ($m_A = m_B = m$). In a 1D elastic collision between two equal masses, the bodies exchange their velocities: bob A comes completely to rest ($v_{Af} = 0$) while bob B moves off with the velocity of A. Since bob A has zero velocity immediately after the collision, **it does not rise at all**.

Practice Question 8 [Single-Correct MCQ — Cosmic Ray Electron vs. Proton Speed Ratio]

In a cosmic ray experiment, an electron is detected with kinetic energy $10\,\mathrm{keV}$ and a proton is detected with kinetic energy $100\,\mathrm{keV}$. Given electron mass $m_e = 9.11 \times 10^{-31}\,\mathrm{kg}$ and proton mass $m_p = 1.67 \times 10^{-27}\,\mathrm{kg}$, which particle is faster and what is the approximate ratio of their speeds (v_e/v_p)?

(A) Proton is faster; $v_e/v_p \approx 0.074$

(B) Electron is faster; $v_e/v_p \approx 13.5$

(C) Electron is faster; $v_e/v_p \approx 183$

(D) Both have the same speed; $v_e/v_p = 1$

Correct Answer: (B)

Worked Solution: From $K = \frac{1}{2}mv^2$, we have $v = \sqrt{\dfrac{2K}{m}}$. Therefore, the ratio of the electron speed to the proton speed is:

$$\frac{v_e}{v_p} = \sqrt{\frac{K_e}{K_p} \times \frac{m_p}{m_e}} = \sqrt{\frac{10\,\mathrm{keV}}{100\,\mathrm{keV}} \times \frac{1.67 \times 10^{-27}\,\mathrm{kg}}{9.11 \times 10^{-31}\,\mathrm{kg}}} = \sqrt{0.1 \times 1833.15} = \sqrt{183.3} \approx 13.54$$

Thus, the electron is faster than the proton by a factor of about 13.5.

Practice Question 9 [Single-Correct MCQ — Gas Molecule Rebounding from a Wall]

A gas molecule hits a rigid stationary horizontal wall with a speed of $200\,\mathrm{m\,s^{-1}}$ at an angle of 30° with the normal, and rebounds with the exact same speed of $200\,\mathrm{m\,s^{-1}}$ at 30° to the normal. Which of the following statements is correct regarding the collision of the molecule-wall system?

(A) Total linear momentum is conserved and the collision is elastic.

(B) Total linear momentum is not conserved, but the collision is elastic.

(C) Total linear momentum is conserved, but the collision is inelastic.

(D) Neither momentum nor kinetic energy is conserved.

Correct Answer: (A)

Worked Solution: Total linear momentum of the isolated (molecule + wall) system is always conserved. The wall receives a normal impulse equal and opposite to the change in momentum of the molecule ($\Delta p = 2mv\cos 30^\circ$), though the wall's recoil velocity is negligibly small because its mass is enormous ($M_{\text{wall}} \gg m$). Since the molecule rebounds with the same speed ($200\,\mathrm{m\,s^{-1}}$) and the massive wall acquires negligible kinetic energy ($\Delta K_{\text{wall}} = \frac{\Delta p^2}{2M_{\text{wall}}} \approx 0$), the total kinetic energy is also conserved; hence the collision is **elastic**.

Practice Question 10 [Assertion–Reason — Conservative Forces & Potential Energy]

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): When a conservative force does positive work on a body, the potential energy of the body decreases.
Reason (R): For a conservative force, the change in potential energy is defined as the negative of the work done by the conservative force ($\Delta V = -W_c$).
In the light of the above statements, choose the correct option:

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (A)

Worked Solution: By definition, $\Delta V = V_f - V_i = -\int_{x_i}^{x_f} F_c\,\mathrm{d}x = -W_c$. Therefore, whenever the conservative force does positive work ($W_c > 0$, such as gravity acting on a falling stone), $\Delta V < 0$, meaning potential energy decreases. Both statements are true and (R) directly explains (A).

Practice Question 11 [Multi-Statement Evaluation — Collisions & Conservation Laws]

Consider the following four statements regarding collisions between two bodies:

1. **Statement I:** In an elastic collision of two bodies, the linear momentum and kinetic energy of *each individual body* are conserved separately.
2. **Statement II:** During the short contact time Δt of an elastic collision between two billiard balls, the total linear momentum of the two-ball system is conserved at every instant, but the total kinetic energy is not constant during contact.
3. **Statement III:** Work done in the motion of a body over a closed loop is zero for every force in nature.
4. **Statement IV:** If the interaction force between two colliding billiard balls depends only on the separation distance between their centres, the collision is necessarily elastic.

Which of the above statements are **correct**?

(A) Statements I and III only

(B) Statements II and IV only

(C) Statements I, II, and IV only

(D) Statements II, III, and IV only

Correct Answer: (B)

Worked Solution:

- **Statement I is False:** It is the *total* momentum and *total* kinetic energy of the two-body system that are conserved, not those of each individual body (energy and momentum are exchanged between the bodies).
- **Statement II is True:** Momentum is conserved at every instant of Δt, whereas kinetic energy is partially converted into elastic deformation potential energy during contact and is only restored after the balls separate.
- **Statement III is False:** Closed-loop work is zero only for *conservative* forces; for non-conservative forces like friction, work over a closed path is negative (non-zero).
- **Statement IV is True:** A force that depends only on separation distance is conservative, so no mechanical energy is dissipated during the collision and recovery, making the collision elastic.

Practice Question 12 [Single-Correct MCQ — Raindrop Terminal Velocity & Resistive Work]

A spherical raindrop of radius $r = 2\,\text{mm}$ falls from a height of $500\,\text{m}$ above the ground. It falls with decreasing acceleration due to viscous air resistance until at half its original height ($250\,\text{m}$) it attains its terminal speed of $10\,\text{m}\,\text{s}^{-1}$ and moves with uniform speed thereafter. What is the work done by the gravitational force in each half of its journey, and what is the total work done by the resistive force over the entire $500\,\text{m}$ journey? (Take density of water $\rho = 10^3\,\text{kg}\,\text{m}^{-3}$ and $g = 9.8\,\text{m}\,\text{s}^{-2}$).

(A) $W_{g,\text{half}} = 0.082\,\text{J}$; $W_r = -0.163\,\text{J}$

(B) $W_{g,\text{half}} = 0.164\,\text{J}$; $W_r = -0.082\,\text{J}$

(C) $W_{g,\text{half}} = 0.082\,\text{J}$; $W_r = -0.082\,\text{J}$

(D) $W_{g,\text{half}} = 0.041\,\text{J}$; $W_r = -0.163\,\text{J}$

Correct Answer: (A)

Worked Solution: Mass of the spherical raindrop ($r = 2 \times 10^{-3}\,\text{m}$):

$$m = \frac{4}{3}\pi r^3 \rho = \frac{4}{3} \times 3.1416 \times (2 \times 10^{-3})^3 \times 10^3 \approx 3.35 \times 10^{-5}\,\text{kg}$$

Since gravitational work $W_g = mgh$ depends only on vertical displacement, the work done by gravity in the first half ($h_1 = 250\,\text{m}$) and second half ($h_2 = 250\,\text{m}$) are equal:

$$W_{g,\text{half}} = (3.35 \times 10^{-5}) \times 9.8 \times 250 \approx 0.082\,\text{J}$$

Total gravitational work over $500\,\text{m}$ is $W_{g,\text{total}} = 2 \times 0.082 = 0.164\,\text{J}$. Final kinetic energy of the drop is:

$$K_f = \frac{1}{2}mv^2 = \frac{1}{2} \times (3.35 \times 10^{-5}) \times 10^2 \approx 0.00168\,\text{J} \approx 1.68 \times 10^{-3}\,\text{J}$$

By the Work-Energy Theorem ($\Delta K = W_{g,\text{total}} + W_r$):

$$W_r = \Delta K - W_{g,\text{total}} = 0.00168\,\text{J} - 0.164\,\text{J} \approx -0.1625\,\text{J} \approx -0.163\,\text{J}$$

11. Common Pitfalls & Rapid Revision Summary

Frequent NEET UG Pitfalls to Avoid

1. **Ambiguous "Work Done" Statements:** Never calculate work without specifying *which force* and *which body* are under consideration. For instance, when lifting a bucket slowly at constant speed, the work done by the lifting force is positive ($+mgh$), the work done by gravity is negative ($-mgh$), and the net work done is zero.
2. **Confusing kWh with Power:** Kilowatt-hour (kWh) is a commercial unit of **energy** ($3.6 \times 10^6\,\text{J}$), whereas kilowatt ($\text{kW}$) is a unit of power.
3. **Applying $\Delta K = 0$ During Collision Contact:** Even in a perfectly elastic collision, kinetic energy is conserved only *after* the collision is complete, not during the brief contact interval Δt when the bodies are deformed.

4. **Changing the Zero of Potential Energy Mid-Problem:** The reference point where $V = 0$ is arbitrary, but once chosen in a problem, you must adhere to it consistently throughout the solution.

Master Formula & Dimensional Summary Table

Physical Quantity	Symbol	Dimensions	SI Unit	Key Defining Equations & Remarks
Work	W	$[\mathrm{ML^2T^{-2}}]$	J	$W = \mathbf{F} \cdot \mathbf{d} = Fd\cos\theta$ (constant force) $W = \int_{x_i}^{x_f} F(x)\,\mathrm{d}x$ (variable force)
Kinetic Energy	K	$[\mathrm{ML^2T^{-2}}]$	J	$K = \frac{1}{2}mv^2 = \frac{p^2}{2m}$; Work-Energy Theorem: $K_f - K_i = W_{\text{net}}$
Potential Energy	$V(x)$	$[\mathrm{ML^2T^{-2}}]$	J	$F(x) = -\dfrac{\mathrm{d}V(x)}{\mathrm{d}x}$; Gravity: $V(h) = mgh$; Spring: $V(x) = \frac{1}{2}kx^2$
Mechanical Energy	E	$[\mathrm{ML^2T^{-2}}]$	J	$E = K + V = \text{constant}$ (if only conservative forces do work) $\Delta E = W_{nc}$ (when non-conservative forces do work)
Spring Constant	k	$[\mathrm{MT^{-2}}]$	$\mathrm{N\,m^{-1}}$	Hooke's law: $F_s = -kx$; $v_{\max} = \sqrt{\frac{k}{m}}\,x_m$
Power	P	$[\mathrm{ML^2T^{-3}}]$	W	$P_{\text{av}} = \frac{W}{t}$; $P_{\text{inst}} = \frac{\mathrm{d}W}{\mathrm{d}t} = \mathbf{F} \cdot \mathbf{v}$; $1\,\text{hp} = 746\,\text{W}$